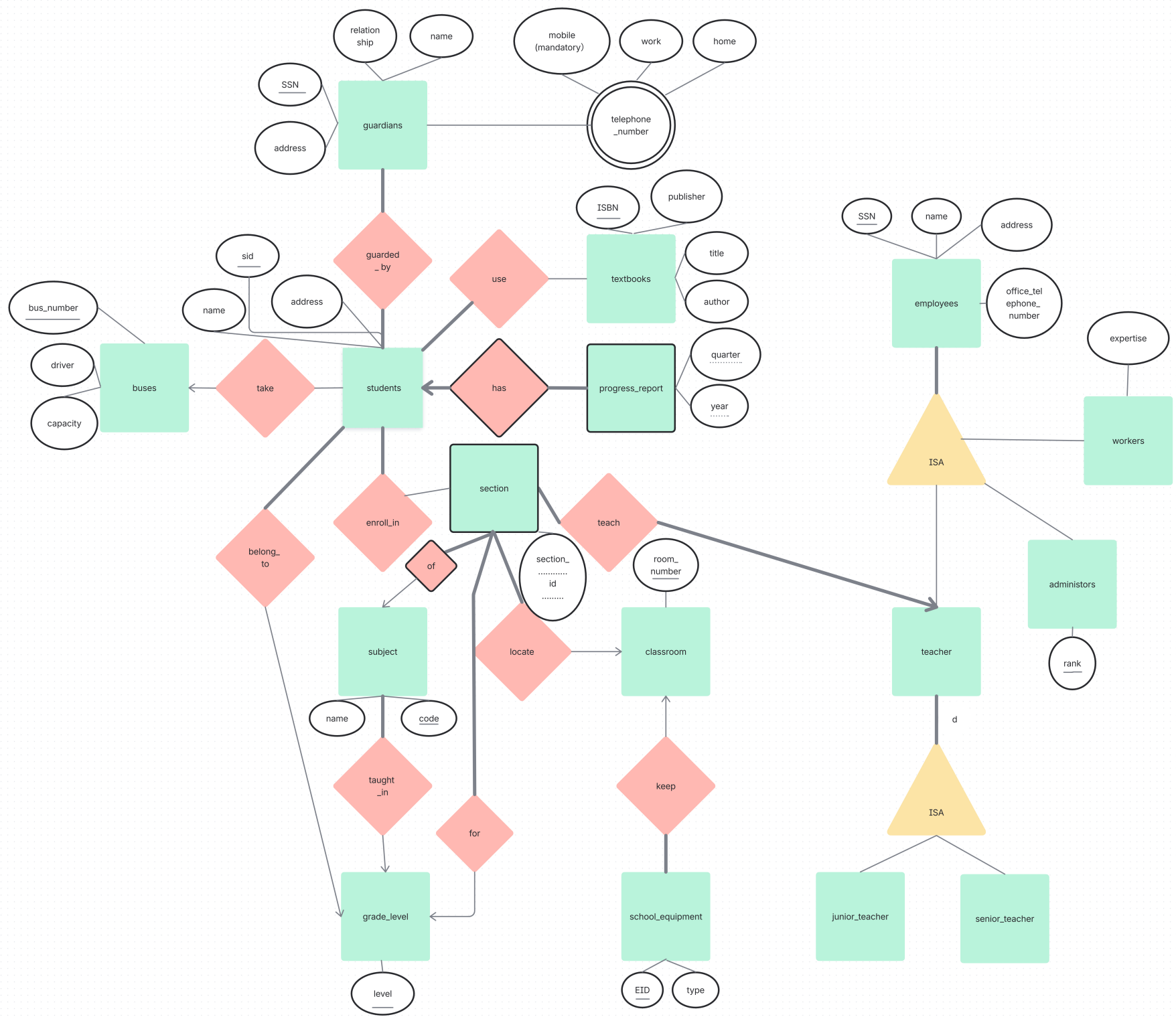




(2)

User requirements that cannot be fully captured by the ERD design

1.Although each Section is associated with a specific GradeLevel in the ERD, enforcing that a student can only enroll in sections of his/her own grade level requires additional integrity constraints beyond standard ERD notation.

2.The ERD models different types of telephone numbers for guardians and indicates that a mobile phone number is mandatory. However, ERD notation cannot strictly enforce that at least one telephone number of type “mobile” must exist; this must be handled at the database or application level.

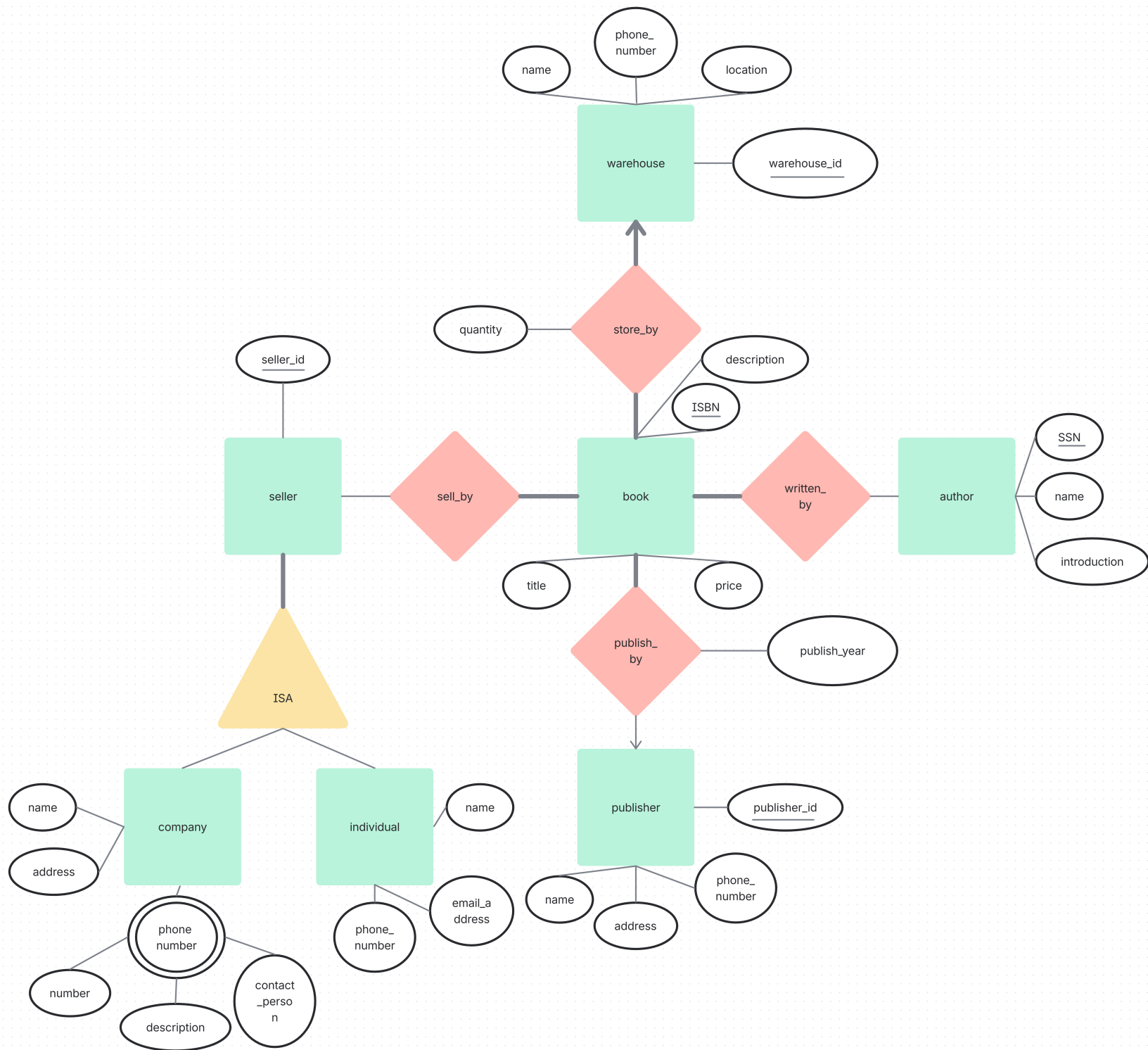
3.Constraints such as administrator ranks being limited to values I-VI, or worker expertise being restricted to specific categories (e.g., IT, Accounting, Design), cannot be fully enforced using ERD alone and require domain or check constraints. ERD does not support CHECK constraints or domain restrictions.

(3)

Name	Type	Attributes	Primary Key	Candidate Key(s)	Partial Key	Related Entity / Owner	ISA
Student	Strong Entity	SID, name, address	SID	{SID}	—	—	—
Guardian	Strong Entity	SSN, name, address, relationship, telephone_number(mobile, work, home)	SSN	{SSN}	—	—	—
Textbook	Strong Entity	ISBN, title, author, publisher	ISBN	{ISBN}	—	—	—
Bus	Strong Entity	bus_number, driver, capacity	bus_number	{bus_number}	—	—	—
Employee	Strong Entity	SSN, name, address, office_telephone_number	SSN	{SSN}	—	—	Employee ISA {Teacher, Worker, Administrator}

Name	Type	Attributes	Primary Key	Candidate Key(s)	Partial Key	Related Entity / Owner	ISA
Teacher	Subclass	SSN	SSN	{SSN}	—	Employee	ISA subclass
Junior_teacher	Subclass	—	SSN	{SSN}	—	Teacher	ISA subclass
Senior_teacher	Subclass	—	SSN	{SSN}	—	Teacher	ISA subclass
Worker	Subclass	Expertise	SSN	{SSN}	—	Employee	ISA subclass
Administrator	Subclass	rank	SSN	{SSN}	—	Employee	ISA subclass
Subject	Strong Entity	code, name	code	{code}	—	—	—
Grade level	Strong Entity	level	level	{level}	—	—	—
Classroom	Strong Entity	room_number	room_number	{room_number}	—	—	—
SchoolEquipment	Strong Entity	EID, type	EID	{EID}	—	—	—
Section	Weak Entity	section_id	(Subject.code, section_id)	{Subject.code, section_id}	section_id	Subject	—
ProgressReport	Weak Entity	year, quarter	(Student.SID, year, quarter)	{Student.SID, year, quarter}	(year, quarter)	Student	—
guarded_by	Identifying Relationship	—	—	—	—	Student–Guardian	—
enroll_in	Relationship	—	—	—	—	Student–Section	—
use	Relationship	—	—	—	—	Student–Textbook	—
take	Relationship	—	—	—	—	Student–Bus	—

Name	Type	Attributes	Primary Key	Candidate Key(s)	Partial Key	Related Entity / Owner	ISA
	ship						
of	Identifying Relationship	—	—	—	—	Section–Subject	—
for	Relationship	—	—	—	—	Section–GradeLevel	—
taught_in	Relationship	—	—	—	—	Subject–GradeLevel	—
teach	Relationship	—	—	—	—	Teacher–Section	—
locate	Relationship	—	—	—	—	Section–Classroom	—
keep	Relationship	—	—	—	—	SchoolEquipment–Classroom	—
Belong_to	Relationship	—	—	—	—	Student–GradeLevel	—
Has	Identifying Relationship	—	—	—	—	Student–progress_report	—



(2)

Although the ERD captures most of the structural requirements, there is one user requirement that cannot be fully represented using an ERD alone:

A contact person whose company sells a book cannot sell the same book as an individual seller at the same time (although he/she may sell other books as an individual seller).

Explanation:

This constraint involves a conditional restriction across an ISA hierarchy (Seller → Company / Individual) and two different relationships (sell_by between Book-Company and Book-Individual). Standard ERD notation does not support expressing such cross-relationship and role-based mutual exclusion constraints.

(3)

Name	Type	Attributes	Primary Key	Candidate Key(s)	Partial Key	Related Entity / Owner	ISA
Book	Strong Entity	ISBN, title, price, description	ISBN	{ISBN}	—	—	—
Author	Strong Entity	SSN, name, introduction	SSN	{SSN}	—	—	—
Publisher	Strong Entity	name, address, phone_number, publisher_id	publisher_id	{publisher_id}	—	—	—
Warehouse	Strong Entity	name, location, phone_number, warehouse_id	warehouse_id	{warehouse_id}	—	—	—
Seller	Strong Entity	seller_id	seller_id	{seller_id}	—	—	Seller ISA {Company, Individual}
Company	Subclass	name, address, phone_number, contact_person	seller_id	seller_id	—	Seller	ISA subclass
Individual	Subclass	name, phone number, email_address	Seller.seller_id	{Seller.seller_id}	—	Seller	ISA subclass
written_by	Relationship	—	—	—	—	Book–Author	—
publish_by	Relationship	publish_year	—	—	—	Book–Publisher	—
store_by	Relationship	quantity	—	—	—	Book–Warehouse	—

Name	Type	Attributes	Primary Key	Candidate Key(s)	Partial Key	Related Entity / Owner	ISA
	hip					ouse	
sell_by	Relationship	—	—	—	—	Book–Seller	—